

Ayush Ghatak

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Profile Summary

Final-year M.Sc. Data Science student with a Computer Science background and hands-on experience in Python, SQL, R, machine learning, and clinical data analysis.

Experience

Advanced Analytics Intern, PharmaQuant Insights Pvt. Ltd.

May 2025 – June 2025

- Synthesized outcomes across 6–8 clinical trials using R (meta, survival packages), applying survival analysis and fixed- and random-effects meta-analysis to evaluate OS, PFS, and ORR (I^2 up to 86%), and generated pooled estimates that were presented to internal stakeholders.
- Used the PICO framework aligned with NCCN clinical guidelines to frame research questions, define comparators, and ensure consistent interpretation of clinical outcomes.

Fixed-Term Research Analyst

June 2022 – July 2022

University of California, Berkeley Haas School of Business

- Processed and analyzed large-scale textual census data using Excel to support qualitative and quantitative research objectives.
- Developed dashboards and analytical reports to enable data-driven decision-making in an academic research lab environment.

Technical Skills

Programming & Tools: Python, R, SQL, C, Power BI, MS Excel.

Data Science & ML: EDA, Statistical Inference, Regression, Machine Learning, Deep Learning, Time Series, NLP, Optimization.

Libraries & Frameworks: pandas, NumPy, scikit-learn, statsmodels, PySpark, XGBoost, tidyverse, matplotlib, seaborn.

Core CS: Data Structures & Algorithms, DBMS, Cryptography.

Education

M.Sc. Data Science, St. Xavier's College (Autonomous), Kolkata — CGPA: 9.01 (till Sem 3) 2024-2026

B.Sc. Computer Science, St. Xavier's College (Autonomous), Kolkata — CGPA: 8.73 2021-2024

Class XII (CBSE) — 94.4% 2021

Class X (CBSE) — 94.2% 2019

Major Projects

Application of Optimization in Drone-Based Photogrammetry

- Built a photogrammetry pipeline on drone imagery using SIFT, FLANN-based KNN, Lowe's ratio test, RANSAC, and triangulation, and applied bundle adjustment with BFGS to reduce reprojection error RMS from 18.71 to 2.91 pixels.
- Optimized GCP alignment by mapping drone positions to nearest GCPs via geodesic distance (pyproj `Geod.inv`), converting EPSG:4326 coordinates to local ENU space, and applying Newton optimization, achieving a 94.6% misalignment error reduction.
- Further minimized overall error by jointly optimizing reconstruction and alignment costs using Fibonacci search, achieving a final combined cost of 0.23 m.

Predictive Modeling of Hospital Readmissions (U.S. Diabetes Dataset)

- Modeled 30-day hospital readmissions using large-scale clinical, demographic, and treatment data, applying data cleaning, feature engineering, and clustered median imputation to address missing values.
- Trained Logistic Regression, Random Forest, and XGBoost models, with XGBoost achieving the best ROC-AUC and identifying key drivers to support reduction of HRRP penalties impacting 83% of U.S. hospitals.

Achievements & Activities

- Secured first place in PowerQuest Hackathon 2025 (HEOR Analytics), PharmaQuant Insights Pvt. Ltd.
- Completed the "Power BI for Beginners" certification on Simplilearn, gaining hands-on experience in data modeling, dashboard design, and interactive data visualization.